Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent

Species Used Hamster, CHO, ovary

Molecules Electroporated DNA: plasmids of 8.4 kb, linearized

Before the Pulse

Cell Growth Medium DMEM, 10% Fetal Bovine Serum, NEAA, Pennicillin /Streptomycin

Growth Phase at Harvest Log

Pre-pulse Incubation 10 min on ice

Wash Solution Berg buffer (see Ref in notes)

The Pulse

Instruments Used Gene Pulser®

Electroporation Temperature 4 °C

Electroporation Medium* Berg buffer (see Ref. in notes)

Cell Density 10(8) cells /0.8ml

Volume of Cells 0.8 ml

DNA Concentration 20 µg

DNA Resuspension Buffer water

Volume of DNA 20 µl

Cuvette Gap 0.4 cm

Voltage 0 .80 kV

Field Strength 2.0 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) Ω none

Time Constant 0 .6 to 0.8 msec

After the Pulse

Outgrowth Medium DMEM, 10% FBS, NEAA, Pen/Strep

Outgrowth Temperature 37 °C

Length of Incubation 2 weeks

Selection Method or Assay Used G418, 8-Aza adenine

Electroporation Efficiency 10 transfectants / µg DNA

Per Cent Survival 20 to 80% (varies)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Ref: Chu,G., Hayakawa, H. and Berg, P. *NAR* **15**, 1131.
Berg buffer: 20 mM HEPES, pH 7.05,
137 mM NaCl, 5 mM KCl, 0.7 mM Na2HPO4, 6 mM dextrose.

Name of Submittor Not given

Institution Address Not given

Survey Number 089